Contents

Chapter 1

Overview

1.1 Symmetries in Physics

What do i even want to do here? I want to give an overview on the role of symmetries in condensed matter and many body physics that is also of interest to computer scientiests, i.e. describe some sort of of system to compute with. However also, there should be an overview about the flavours of (topological) phases, orders and entanglements. Most important foremost would be a paragraph on what "symmetry" as a concept refers to. Then we would want to classify the symmetries we can (or want to!) adress with tensor networks (and string (spin?) networks!). Landau theory and it's failure would be a great starting point for a condensed matter approach, this is directly correlated with all of topological order, symmetric hamiltonians and dualities.

So for now im thinking to treat this as a collection of independent chapters. First on global symmetries in tensor networks. Work backwards from the singh papers (however these are actually post-anyon networks (c.f. [Pfe+10])) and establish the extent that these papers deal with.

Then write one section on the anyonic networks with sparse mentions of symmetries. Of course we could give representations as a special case here if we take the representations to form an anyon theory (really just a fusion category) as given in [Sim23, § 20] but I do not know if this is desired. Well, the general theory is handled in [DH25] with utmost care, however I would like to give some duly needed proofs (in really all of these papers, I believe the word *proof* only appears once in all of them!) i.e. as done already with our decomposition theorem 2.1.5.

And then it really is just open ended. I think a nice description of dualities is duly managable but this of course requires the discussion of module categories as in [LDV24a] as well as a restriction on the type of systems we deal with. In fact, [BLV23] gives us a version of schurs lemma for bimodule categories and mentions its application as clebsch gordan tensors.

Another option would be to explore MPO-symmetries, as these share quite a lot of language with our decompositions (in fact, the symmetries they generate are a direct superset of ours!). There is also the consideration of virtual only symmetries, and I currently remember a description of them here [dGro22]. Both the MPO-case and the virtual symmetry case require a discussion of projective symmetries as well.

Someting I currently know nothing about is the theory on *symmetry protected topological* (SPT) phases, but it does sound relevant. [dGro22] also gives some tensor network methods in this context.

The topic of topological order would require a discussion of string nets, and inthereof the toric code. I need to understand this anyways, so why not write it down nicely?

On a side note, all of this is an instance of penrose notation [Pen71], and im not yet sure what the difference between spin and string diagrams are. Also, structures such as the metric Lie Algebra TTS given in [nLa26] seem to admit a very similar calculus, but not with defined braiding as such. Also, there is the topic of categorical computationl using finite groups (but this also seems to be similar to the formalism in Simon TQ), accounts are in the work of Mochon [Moc03][Moc04] and i remember something about abelian groups not being suitable in a paper of Bravyi and Koening (find this). And of course, categorical computation in general although i do not believe this matters here [KZ23].

1.2 Review: Landau Theory and it's limitations

All matter is made of (what exactly? 3 fundamentals), but can display many different structural properties called *phases*, such as liquid, crystal, solid, (what else? condensations? is there a nice diagram?). These arrangements of matter can be manipulated, should the structural property change we say that a *phase transition* has occured.

Landaus *symmetry breaking* theory [<**empty citation**>] was thought to classify all phases of matter (What are the 230 Groups/Crystals here?). However, as shown by [Wen95], there are phases that are not described by spontaneous symmetry breaking but rather a new quantity, the dependence on the topology of the ground state.

There is a nice review here [Wen13]. There is also a paper by Xiao-Gan Wen from 2020, find this.

1.3 Introduction

1.3.1 Some basic Category Theory

Here we introduce some basic category concepts. For a review of categories for quantum systems see [Sel11], for a use in tensor networks see [VVW18] or [Bia20](Check this!).

Similarly to the concepts above, we introduce categories because there are some that mirror the graphical language used by tensor networks exactly. Furthermore, both our later treatment of anyons and symmetries in Chapters .... makes use of these concepts. Here we mostly follow the language used by [Vak25].

Definition 1.3.1 (Category)**.** A category C is built by a collection of of objects, which we usually denote by $obj(\mathrm{C})$, and a set of morphisms $\mathrm{Hom}_C(A, B)$ for each pair A, B of objects, where we refer to A as the domain and B as the codomain. If we want to refer to an object in C, we usually just write $A \in \mathrm{C}$.

Furthermore, there is a binary operator $\circ : Hom_C(A, B) \times Hom_C(B, C) \to Hom(A, C)$ such that this is associative. Also, we have an identity morphism for each object A denoted by id_A s.t. $id_A \circ f = f$ and $g \circ id_B = g$ for all appropriate morphisms.

We say there is an *isomorphism* between two objects if there exists an inverse morphism (Just copy vakil here.).

As an example $Hilb_{fd}$ forms a category where the objects are f.d. Hilber spaces and the morphisms are the linear maps between them.

Realize that this encodes all the needed information about sequential composition.

Definition 1.3.2 (Functor)**.** A *Functor* is a map from category A to category B and is given by the data

$$F : obj(\mathrm{A}) \to obj(\mathrm{B}) \quad \text{and} \quad F : Hom(A, B) \to Hom(F(A), F(B))$$

such that F preserves the identity and composition

$$F(id_A) = id_{F(A)} \quad \text{and} \quad F(f \circ g) = F(f) \circ F(g)$$

Now, in a quantum system we want the option to also describe *parallel composition.* If we have two systems A and B we want to apply processes f and g on the combined system $A \otimes B$ as $f \otimes g$. So we introduce extra structure to our categories in the form of a functor $- \otimes -$. This acts between

$$\otimes : \mathrm{C} \times \mathrm{C} \to C$$

where

$$\otimes : obj(\mathrm{C}) \times obj(\mathrm{C}) \to \mathrm{C} : (a, b) \mapsto a \otimes b$$

and

$$\otimes : Mor(A, A') \times Hom_{\mathrm{C}}(B, B') \to Hom_{\mathrm{C}}(A \otimes A', B \otimes B'), (f, g) \mapsto f \otimes g.$$

Now, the tensor product in general is not associative, i.e. the spaces $A \otimes (B \otimes C)$ and $(A \otimes B) \otimes C$ are only isomporphic. So we further define a natural isomorphism

$$\alpha : (A \otimes B) \otimes C \to A \otimes (B \otimes C)$$

called the associator.

We can now make this construction a formal definition:

Definition 1.3.3 (Monoidal Category)**.** A monoidal category is a category C equipped with

1. A bifunctor $\otimes : C \times C \to C$.
2. A unit element $1 \in C$
3. An isomorphism
$$\alpha : - \otimes (- \otimes -) \to (- \otimes -) \otimes -$$
4. Left and right unitors $\gamma : 1 \otimes - \to -$ and $\delta : - \otimes 1 \to -$.

We denote this by the triple $(C, \otimes, 1)$ where we omit the unitors. Otherwise this is a 5 or 6 tuple TODO.

TODO: graphical calculus.

1.3.2 Tensors (Everything here is $fdVect_k$)

Lets list some definitions of Tensor. A Tensor T can be regarded as an element of a tensor product space

$$T \in \bigotimes_{j \in J} V^j$$

and we say T is a $|J|$-legged tensor.

Given finite dimensional Vector spaces V and W, we have the identity that $V \otimes W^* \equiv Hom(W, V)$. Now, if we have a Tensor $T \in V_1 \otimes \cdots \otimes V_n$, and since for any Vector space V, $(V^*)^* = V$, we can peel off any number of V_is, take their duals and represent T as a map

$$T : V_n^* \otimes \cdots \otimes V_{k+1}^* \to V_1 \otimes \cdots \otimes V_k$$

Note that we present the domain here in a contravariant fashion due to the contravariance of the dual.

Lets look at an example, let $t : V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. We can represent as an operator in dirac notation:

$$t = \sum_{i_1, i_2, j_1, j_2} T_{j1j2i1i2} |v_{i_1}\rangle |v_{i_2}\rangle \langle w_{j_1}| \langle w_{j_2}|$$

Graphically, we represent a general tensor

$$T : (W_n^* \otimes \cdots \otimes W_1) \to (V_1 \otimes V_2 \otimes \cdots \otimes V_n)$$

as follows, where we choose incoming arrows for duals in the domain and outgoings for normal spaces, and on the bottom we choose incoming for duals and outgoing for normals.

1.3.3 (Algebraic?) Theory of Anyons

1.3.4 Symmetric Tensors

Let G be a compact group. Consider a tensor T living in the tensor product space of its legs:

$$T \in V_1 \otimes \cdots \otimes V_n \otimes W_m^* \otimes \cdots \otimes W_1^*$$

Let $\rho_i : G \to \mathrm{GL}(V_i)$ be the linear representation of G on each vector space V_i, and let $\sigma_j^* : G \to \mathrm{GL}(W_j^*)$ be the dual representation on W_j^*.

The total action of the group G on the entire tensor product space is given by

$$\Pi : G \to \mathrm{GL}\left(\bigotimes_i V_i \otimes \bigotimes_j W_j^*\right)$$
$$\Pi(g) = \rho_1(g) \otimes \cdots \otimes \rho_n(g) \otimes \sigma_m^*(g) \otimes \cdots \otimes \sigma_1^*(g)$$

A tensor T is defined to be a **symmetric tensor** (or G-invariant tensor) if it is invariant under simultaneous action on all legs for all $g \in G$ [SPV10]:

$$\Pi(g)T = T \quad \forall g \in G$$

By interpreting T as a map

$$T : W_1 \otimes \cdots \otimes W_m \to V_1 \otimes \cdots \otimes V_n$$

it follows directly from the definition of global invariance that the tensor must commute with the group action and we get the following definition of a symmetric tensor:

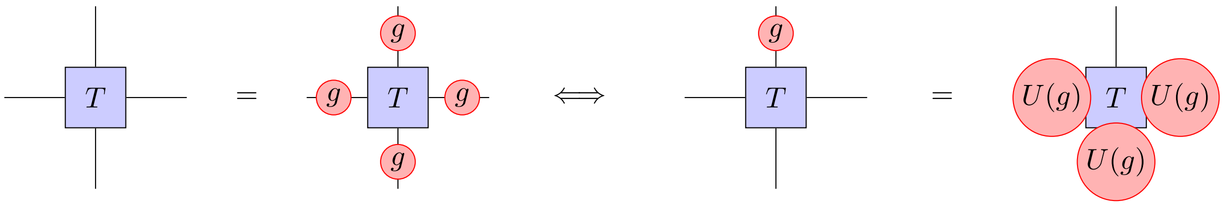
Definition 1.3.4 (symmetric tensor)**.** A *symmetric tensor* is an intertwiner of the representations of the group G:

$$\bigotimes_{j=1}^{m} \sigma_j(g)T = T \bigotimes_{i=1}^{n} \rho_i(g) \quad \forall g \in G,$$

which we denote by

$$T \in Hom_G(W, V).$$

TODO: what is the definition of an MPO representation? This could include this as well, maybe as $Hom_{MPO}(W, V)$.



Corollary 1.3.4.1 (Networks of symmetric tensors are symmetric)**.** Since a tensor network is built by composition and application of the $(co)ev$ maps, both of which are intertwiners, the full network is also an intertwiner. Important there is that this entire discussion is only about FdVect or FdHilb for now, but i think the same applies to any braided category.

1.3.5 Some Important Facts about Hamiltonians

Definition 1.3.5 (Gapped Hamiltonian)**.** Let H be a hamiltonian acting on a finite system of size L. Let

$$E_0 \leq E_1 \leq \cdots$$

be the eigenvalues. The spectral gap is then

$$\Delta(L) = E_1 - E_0.$$

The Hamiltonian is gapped if there is a constant Δ

1.3.6 Review of Symmetric Tensor Networks.

Generally, we follow the notations of [SV12], [Sch+20], [Sch+20], [Sch20] which gave an overview for implementing $SU(2)$ invariant tensor networks. A general overview for (Non)abelian symmetries is [SPV10], and a detailed account for the abelian $U(1)$ in [SPV11]. There are a multitude of implementations given in major packages, such as ITensor, [FWS22], Cyten, TenPy, [Hau+24], QSpace, [Wei24] ChemTensor, UniTensor, YasTn (Abelian Only) [Ram+25] (MPSKit?), (Cytnx (Abelian only i think.))). A general account of symmetries is given in [Unf24] with its respective implementation in [Hau+24] (We have to build cytens citation manually.)

The motivation for constructing symmetric tensor networks is as follows: Assume we have a (unitary) representation of a group G denoted $U(g)$. Furthermore assume we have a hamiltonian that commutes with $[U(g)^{\otimes L}, H] = 0$. If the ground state is unique, we can obtain

$$U(g)^{\otimes L} H \left|\psi_0\right\rangle = H(U(g)^{\otimes L}) \left|\psi_0\right\rangle = E_0 U(g)^{\otimes L} \left|\psi_0\right\rangle$$

and thus $U(g)\left|\psi_0\right\rangle$ is a ground state as well. By uniqueness

$$U(g)\left|\psi_0\right\rangle = e^{i\theta(g)}\left|\psi_0\right\rangle$$

../experiments/plots/su2_block_structure.pdf

Figure 1.1: Three different presentations of the XXZ Hamiltonian. (i): In the default computational basis of spin vectors. (ii) In a reordering of the computational basis into equal spin up (down) classes, i.e. no spin up, one spin up etc..., here U is simply a permutation matrix. (iii) After changing the basis into the S^2 basis.

Definition 1.3.6 (XXZ-Hamiltonian)**.** The XXZ Hamiltonian on N sites is

$$H = J \cdot \sum_{i=1}^{N} (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_i^z). \tag{1.1}$$

As given in 1.3.6, this acts on the *spin basis* made of states $\left|s_1 s_2 \dots s_n\right\rangle$ where $s_i \in \{\uparrow, \downarrow\}$.

We can plot H and realize the sparsity structure thereof. Furthermore, a suitable rearrangement induces a block structure 1.1.

Now, we can realize about this hamiltonian is that there is a symmetry with the representations of $U(1)$. Lets review this shortly. A representation of $U(1)$ is a group homomorphism

$$\rho : U(1) \to GL(V), \quad \rho(e^{i\theta_1})\rho(e^{i\theta_2}) = \rho(e^{i(\theta_1+\theta_2)}).$$

The XXZ model lives in the Hilbert Space $\mathcal{H} = (C^2)^n$ and the relevant representation is generated by the total magnetization operator

$$S^z_{\text{tot}} = \frac{1}{2}\sum_{i=1}^{N} \sigma_i^z$$

The group action is given by

$$U(\theta) = e^{-i\theta S^z_{\text{tot}}}.$$

We now claim that H is invariant under $U(\theta)$.

$$U(\theta)HU(\theta)^\dagger = H, \qquad \forall\theta \in \mathbb{R}.$$

Equivalently,

$$[H, S^z_{\text{tot}}] = 0.$$

Recall that when two operators commute, we can *simultaneously diagonalize* them [HJ12, Theorem 1.3.12].

$$[H, S^z_{\text{tot}}] = \sum_i^N [\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y + \Delta\sigma_i^z\sigma_i^z, \sum_{j=1}^N \sigma_j^z]$$

and since operators on different sites commute, only $j = i$ and $j = i+1$ contribute. So we ought to write

$$\sum_i^N [\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y + \Delta\sigma_i^z\sigma_i^z, \sigma_i^z + \sigma_{i+1}^z]$$

and now using the identity $[AB, C] = A[B, C] + [A, C]B$:

$$[\sigma_i^x\sigma_{i+1}^x, \sigma_i^z + \sigma_{i+1}^z] = -2i(\sigma_i^y\sigma_{i+1}^x + \sigma_i^x\sigma_{i+1}^y)$$

$$[\sigma_i^y\sigma_{i+1}^y, \sigma_i^z + \sigma_{i+1}^z] = 2i(\sigma_i^x\sigma_{i+1}^y + \sigma_i^y\sigma_{i+1}^x)$$

which cancel out. Furtermore, the z spins commute as well. Thus the whole sum is 0 and we are done.

In the $SU(2)$ case, we have the generators

$$S^\alpha_{tot} = \frac{1}{2}\sum_{i=1}^N \sigma_i^\alpha, \quad \alpha \in \{x, y, z\}$$

The representation here is $U(R) = e^{i\theta n\cdot S_{tot}}$ and the Hamiltonian is invariant under

$$U(R)HU(R)^\dagger = H, \quad R \in SU(2)$$

Since

$$[S^x_{tot}, S^y_{tot}] = iS^z_{tot}$$

we need a new operator for the simulatenous diagonalization

$$S_{tot} = \sum_{a\in\{x,y,z\}} (S^a_{tot})^2$$

and we need to verify this is indeed the correct operator TODO.

The eigenvalues of S^2_{tot} are $S(S+1)$ where S are the half integers.

Also we have that

$$(\mathbb{C}^2)^{\otimes 6} \cong V_{\frac{1}{2}}^{\otimes 6} \cong 5V_0 \oplus 9V_1 \oplus 5V_2 \oplus V_3$$

So we expect 20 blocks? 5 have dimension 1, 9 have dimension 3, 5 have dimension 5 and 1 has 7.

MPS

Definition 1.3.7 (Matrix Product State)**.** A *Matrix Product State* is a quantum state $|\psi\rangle$ with n legs written as the contraction of n trivalent tensors $\{A^{[i]}\}_{i\in[n]}$. Formally this reads

$$|\psi[A]\rangle = \sum_{i_1,\ldots,i_n} \mathrm{Tr}\Big(A^{[1]}_{i_1} A^{[2]}_{i_2} \cdots A^{[n]}_{i_n}\Big) |i_1 i_2 \cdots i_n\rangle ,$$

where $A^{[i]}_{i_k} \in \mathbb{C}^{D_i \times D_{i+1}}$. The elements of the collection $\{D_i\}_{i\in[n]}$ is referred to as the *bond-dimensions*.

Note that this is a state with closed boundaries. If we want open boundaries we have to replace the trace with appropriate end vectors (nice source?).

Example 1. We can represent each product state $|\Psi\rangle = |\psi\rangle^{\otimes n}$ as a MPS with bond dimension 1.

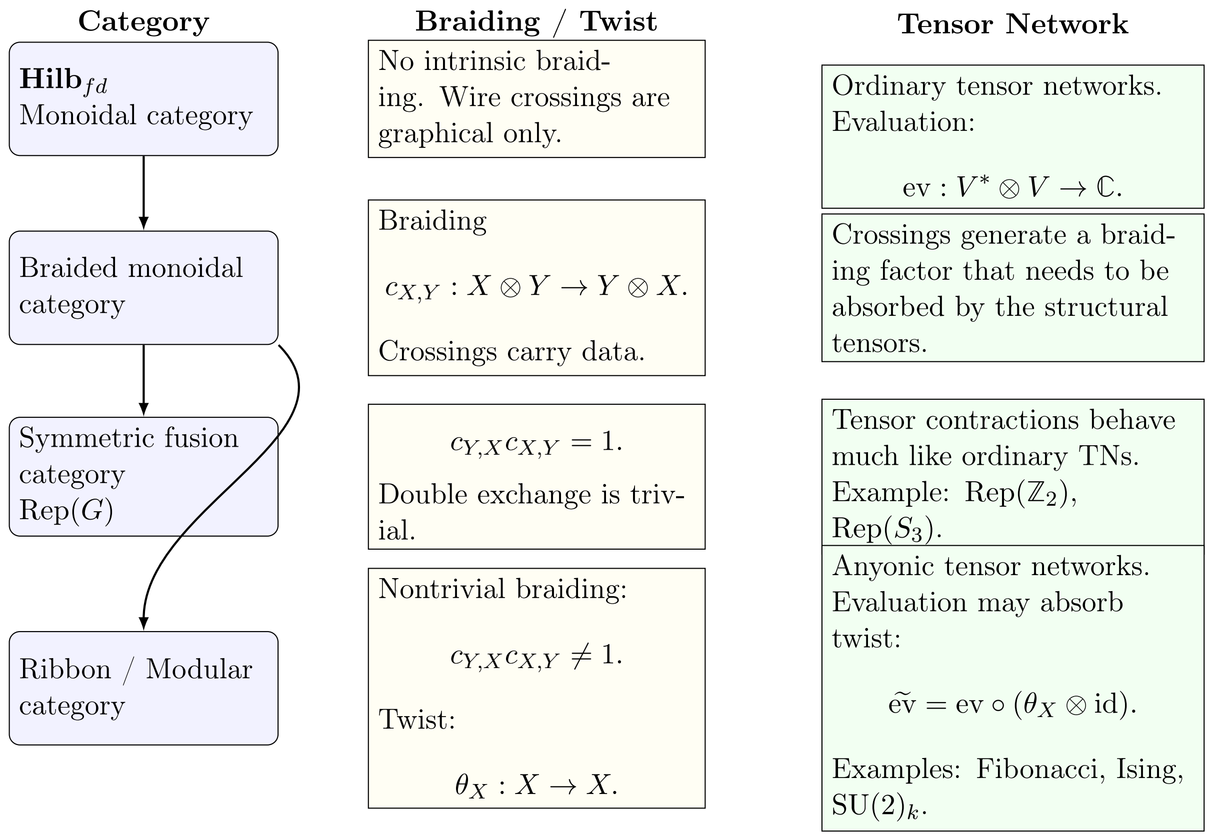
Theorem 1.3.8 (Fundamental Theorem of MPS)**.** If two MPS states generated by A and B respectively. Figure out what we can do with this.

Proof [Cir+17],[VVW18], [Per+08].

Chapter 2

Decompositions for Categorical Networks

This chapter generalizes the decomposition in [SV12, Equation 27] to both Fusion Categories and the Representation categories of compact groups. Note that in [DH25], such a generalization is given for representations of finite groups, which does not include $SU(2)$. However I do not yet know if we should do so, since I not yet sure how to argue the F moves here. If we instead take the deformation $U_q(sl_2)$ we get finiteness (and fusion cats?), unfortunately $U_q(sl_2)$ is not a group.



2.1 Decomposition of semisimple Categories

Lemma 2.1.1. For a vector space W, there is an isomorphism

$$\bigoplus^m W \simeq \mathbb{C}^m \otimes W.$$

Proof. Let $e_{i i \in [m]}$ be the standard basis on $\mathbb{C}^m$. Let

$$\Phi : \mathbb{C}^m \otimes W \to \oplus^m W$$

be defined by

$$\phi(e_i \otimes w) = (0, \ldots, w, \ldots 0)$$

and linear, thus

$$\Phi\left(\sum_i c_i e_i \otimes w_i\right) = (c_1 w_1, \ldots, c_m w_m).$$

is clearly bijective, and we have our desired isomorphism. □

Lemma 2.1.2. Let D be a finite dimensional k-vector space and let V be an object of a k-linear category C. Then

$$D \otimes V \cong V^{\oplus \dim D}$$

Proof. Let $n = \dim D$. Take a basis $\{e_i\}$ for D. This yields an isomorphism

$$D \cong \bigoplus_{i=1}^{n} k \cdot e_i \cong k^n$$

Since in a k-linear category the tensor product is additive (should we prove this?), we can distribute

$$D \otimes W \cong \bigoplus_{1}^{n} (k \cdot e_i \otimes W) \cong \bigoplus W = W^{\oplus n}$$

□

Lemma 2.1.3 (Isotopic Decomposition)**.** Let C be a semisimple k-linear category with simple objects $\{R_a\}$. Then every object $V \in C$ admits a canonical decomposition

$$V \cong \bigoplus_a D_a \otimes R_a. \tag{2.1}$$

where we refer to $D_a \coloneqq \mathrm{Hom}_C(R_a, V)$ as the *degeneracy* of the space.

Proof. Since C is semisimple, there exist integers $m_a \geq 0$ such that

$$V \cong \bigoplus_{a \in L} \bigoplus_{\_=1}^{D_a^V} a \cong \bigoplus_{a \in L} R_a^{\oplus m_a}.$$

Applying the additive functor $\mathrm{Hom}_C(R_b, -)$ yields

$$\mathrm{Hom}_C(R_b, V) \cong \bigoplus_{a \in L} \mathrm{Hom}_C(R_b, R_a^{\oplus m_a}) \cong \bigoplus_{a \in L} \mathrm{Hom}_C(R_b, R_a)^{\oplus m_a}.$$

Then, by Schurs lemma

$$\bigoplus_{a \in L} Hom_C(R_b, R_a)^{\oplus m_a} = k^{\oplus m_b}$$

□

So we actualy need this: Maybe this is simpler if we always let the degeneracy be C?

Lemma 2.1.4. Let C be a k-linear category and let D, E be finite-dimensional k-vector spaces. For any objects $R, S \in C$, there is a natural isomorphism

$$\mathrm{Hom}_C(D \otimes R, E \otimes S) \cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. Proof by isomorphism chains:

$$\begin{aligned}
\mathrm{Hom}_C(D \otimes R, E \otimes S) &\cong \mathrm{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by 2.1.1} \\
&\cong \bigoplus_{i=1}^{m} \bigoplus_{j=1}^{n} \mathrm{Hom}_C(R, S) && \text{by additivity of } \mathrm{Hom}_C \\
&\cong k^{mn} \otimes \mathrm{Hom}_C(R, S) && \text{by 2.1.1} \\
&\cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S) && \text{since } \mathrm{Hom}_k(D, E) \cong k^{mn}
\end{aligned}$$

(The last as s) □

Theorem 2.1.5 (Decomposition of Semisimple Categories)**.** Let C be a semisimple, monoidal, k-linear category whose tensor product is k-linear in each variable, with simple objects $\{R_a\}_{a\in L}$ and let V_i, W_i be a collection of objects in C. Every morphism

$$T \in \mathrm{Hom}_C\left(\bigotimes_{i=1}^{n} V_i, \bigotimes_{i=1}^{m} W_i\right)$$

admit a decomposition as

$$T \in \bigoplus_{\hat{a},\hat{b}} P^{\hat{a},\hat{b}} \otimes Hom_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \ldots, a_n)$ and P is a k-vector space of dimension ???

Proof.

$$\begin{aligned}
&\mathrm{Hom}_C\left(\bigotimes_i V_i, \bigotimes_j W_j\right) \\
&\quad\cong \mathrm{Hom}_C\left(\bigotimes_i\Big(\bigoplus_{a_i} D_{i,a_i}\otimes R_{a_i}\Big), \bigotimes_j\Big(\bigoplus_{b_j} E_{j,b_j}\otimes R_{b_j}\Big)\right) && \text{by 2.1.3} \\
&\quad\cong \mathrm{Hom}_C\left(\bigoplus_{\hat{a}} D_{\hat{a}}\otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}}\otimes R_{\hat{b}}\right) && \text{since } \otimes \text{ is } k\text{-linear in each variable} \\
&\quad\cong \bigoplus_{\hat{a},\hat{b}} \mathrm{Hom}_C\big(D_{\hat{a}}\otimes R_{\hat{a}}, E_{\hat{b}}\otimes R_{\hat{b}}\big) && \text{by additivity of } \mathrm{Hom}_C \\
&\quad\cong \bigoplus_{\hat{a},\hat{b}} \mathrm{Hom}\big(D_{\hat{a}}, E_{\hat{b}}\big)\otimes \mathrm{Hom}_C\big(R_{\hat{a}}, R_{\hat{b}}\big) && \text{by 2.1.4.}
\end{aligned}$$

□

Corollary 2.1.5.1 (Operators are block matrices)**.** This is also given here: [KZ22, Remark 3.4.8] An operator

$$Hom(V, W) \cong Hom_C(\bigoplus_a R_a^{\oplus d_a}, \bigoplus_b R_b^{\oplus e_b}) \cong \bigoplus_{a,b} Hom(D_a, E_b) \otimes Hom_C(R_a, R_b)$$

and by Schurs lemma this is just

$$\bigoplus_{a\in Irr(V)\cap Irr(W)} Hom(D_a, E_a) \cong \bigoplus_{a\in Irr(V)\cap Irr(W)} Mat_{d_a\times e_a}(k)$$

Corollary 2.1.5.2. Let $Q_{\mu,\eta}$ be a basis of $Hom_C\big(\bigotimes_{\hat{a}} R_{\hat{a}}, \bigotimes_{\hat{b}} R_{\hat{b}}\big)$ where μ denotes the multiplicities and η the internal fusions. In a *multiplicity-free* Category we will omit μ as a label.

Example 2. If we let our monoidal semisimple category just be $\mathrm{Vec}_{\mathbb{C}}$, the only simple object is $\mathbb{C}$. Thus there is also only the trivial biproduct over one sector.

Example 3. If we restrict to the category $\mathrm{Rep}(SU(2))$ we can build the symmetric tensor networks as given in [SV12]. TODO: actual ly do this and convince yourself that the notations amd the triplet basis workd as desired.

Example 4. Take a 2 in 1 out tensor in $\mathrm{Rep}(SU(2))$ as

$$T \in Hom_{\mathrm{Rep}(SU(2))}(V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}, V_{\frac{1}{2}})$$

Since all parts are simple, nothing happens.

Example 5. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$
\begin{aligned}
&Hom_C(\bigoplus_{a,b,c\in(0,1)} (C \otimes C \otimes C) \otimes V_a \otimes V_b \otimes V_c, \bigoplus_{b\in(0,1)} C \otimes V_b) \\
&= \bigoplus_{a,b,c,d\in(0,1)} Hom_C((C \otimes C \otimes C) \otimes (V_a \otimes V_b \otimes V_c), C \otimes V_d) \\
&= \bigoplus_{a,b,c,d} Hom(C^3, C) \otimes Hom_{\mathrm{Rep}(SU(2))}(V_a \otimes V_b \otimes V_c, V_d)
\end{aligned}
$$

Example 6. Easier. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$
\begin{aligned}
&Hom_C\left(\bigoplus_{a,b\in(0,1)} (C \otimes C) \otimes V_a \otimes V_b \bigoplus_{b\in(0,1)} C \otimes V_b\right) \\
&= \bigoplus_{a,b,c\in(0,1)} Hom_C((C \otimes C) \otimes (V_a \otimes V_b), C \otimes V_c) \\
&= \bigoplus_{a,b,c} Hom(C^2, C) \otimes Hom_{\mathrm{Rep}(SU(2))}(V_a \otimes V_b, V_c)
\end{aligned}
$$

Example 7. Consider the same operator as in example 5 but with its incoming legs fused together. So we have $V_{in} = V_{\frac{1}{2}}^4$ and $V_{out} = V_{\frac{1}{2}}^2$. This decomposes $V_2 \oplus V_1^{\oplus 3} \oplus V_0^{\oplus 2}$. Then we can directly use 2.1.5.1 to obtain

$$Hom(V_{in}, V_{out}) =$$

Example 8. Lets look at a tensor with in legs $(\tau \otimes \tau)$ and τ and out legs τ.

$$Hom_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) = Hom_C()$$

$$
\begin{aligned}
&\mathrm{Hom}_C\big(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau),\ k \otimes \tau\big) \\
&\cong \mathrm{Hom}_C\big((k \otimes 1 \otimes k \otimes \tau) \oplus (k \otimes \tau \otimes k \otimes \tau),\ k \otimes \tau\big) \\
&\cong \mathrm{Hom}_C(k \otimes 1 \otimes k \otimes \tau,\ k \otimes \tau) \oplus \mathrm{Hom}_C(k \otimes \tau \otimes (k \otimes \tau),\ k \otimes \tau) \\
&\cong \mathrm{Hom}(k \otimes k, k) \otimes \mathrm{Hom}_C(1 \otimes \tau, \tau) \oplus \mathrm{Hom}(k \otimes k, k) \otimes \mathrm{Hom}_C(\tau \otimes \tau, \tau).
\end{aligned}
$$

Example 9 (Anyonic matrix operator)**.** Finally, consider a $4 \to 4$ anyon operator with one leg on each side, representing the space $\mathrm{Hom}((\tau \otimes \tau \otimes \tau \otimes \tau), (\tau \otimes \tau \otimes \tau \otimes \tau))$. THe full decomposition is $2 + 3\tau$. Hence we obtain

$$\mathrm{Hom}_C\big((k^2 \otimes 1) \oplus (k^3 \otimes \tau), (k^2 \otimes 1) \oplus (k^3 \otimes \tau)\big)\,.$$

$$= \bigoplus_{a,b\in(1,\tau)} \mathrm{Hom}_C(k^{m_a} \otimes a,\ k^{m_b} \otimes b) = \bigoplus_{a,b\in(1,\tau)} \mathrm{Hom}(k^{m_a}, k^{m_b}) \otimes Hom_C(a, b)$$

where

$$m_1 = 2, \qquad m_\tau = 3.$$

For this case, Schurs lemma applies. So we have

$$= Hom(k^2, k^2) \oplus Hom(k^3, k^3)$$

which is just a block matrix with indexing $(1_1, 1_2, \tau_1, \tau_2, \tau_3)$.

Example 10 ($U(1)$ Symmetry)**.** Technically we would want to show here that this is simpler in general. TOOD. Make a theorem? Or a corollary?

Example 11 (Ground state on 4 anyons)**.** We can be given a ground state in the operator form as $Hom(\tau^4, \tau)$.

2.1.1 Q intertwiners

Lets look at 2 paths for the 4 index tensor, $(a, b, c \to d)$ and $(a, b \to c, d)$.

$$(Q^{j_e}_{j_a j_b j_c})_{m_1,m_2,m_3,m_4} = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e\rangle \langle j_e m_e; j_3 m_3 | j_4 m_4\rangle$$

while the other topology would look like

$$Q \ldots = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e\rangle \langle j_e m_e | j_3 m_3; j_4 m_4\rangle$$

and as an operator this would have to live in $Hom(V_{j_1} \otimes V_{j_2}, V_{j_3} \otimes V_{j_4})$

The yoga tree $(-1, 1, -3), (-2, 1, -4)$ Gives us a good mixture of those. This Q would then have to look like

$$Q^1_{-1-2-3-4} = \sum_1 \langle j_{-1} m_{-1}; j_1 m_1 | j_{-3} m_{-3}\rangle \langle j_{-2} m_{-2} | j_1 m_1; j_{-4} m_{-4}\rangle = \sum_{m_1} C_{-1,1\to-3} \cdot C_{-2,1\to-4}$$

How would we do this when we have a contraction of two simultaneous indices:

$$Q^{1,2}_{-1-2} = \sum_{1,2} \langle -1|1; 2\rangle \langle 1; 2|-2\rangle$$

If we define τ to be the fusion trees topology, then we can define ${}_\tau Q^i_j$

What would this look like for general symmetry groups G? Lets choose a basis of its invariant decomp $|\lambda_a t_a \mu_a\rangle$. Then the generalized CG coefficient would look like

$$(C^\alpha_{\lambda_a\lambda_b\to\lambda_c})^{\mu_c}_{\mu_a\mu_b} = \langle \lambda_a\mu_a; \lambda_b\mu_b|\lambda_c\mu_c\rangle_\alpha$$

where alpha denotes the singular fusion multiplicity at μ_c.

Now to fully specify Q here for k sectors (λ_i, μ_i), we need to give it $k-3$ internal edges $(\lambda_{e_1}, \ldots, \lambda_{e_{k-3}})$ along with $k-2$ mulitplicity symbols, $(\alpha_1, \ldots, \alpha_{k-2})$

Can we write down a general formula for Q?

This would then be

$$Q_\lambda$$

2.2 Categories

2.2.1 Tensor and Fusion Categories

Let C be a category with bifunctor

$$\otimes : C \times C \to C, \quad (X, Y) \to X \otimes Y.$$

and the isomporphism $F : (- \otimes -) \otimes - \simeq - \otimes (- \otimes -)$

... introduce fusion categories here, with associativity and commutativity restrictions, I think this would be a good start.

Definition 2.2.1 (Monoidal Category)**.** A monoidal category is a category C with a bifunctor $\otimes : C \times C \to C$

Let C and D be linear abelian categories. Then the *Deligne Tensor Product* $C \boxtimes D$ is a linear abelian category with a bifunctor

$$\boxtimes : C \times D \to C \boxtimes D$$

with morphisms given by

$$Hom_{C\boxtimes D}(X \boxtimes Y, W \boxtimes V) = Hom_C(X, W) \otimes Hom_D(Y, V)$$

Definition 2.2.2 (The Deligne Tensor Product of Categories)**.** Let k be a field, and let $\mathcal{C}$ and $\mathcal{D}$ be two k-linear, finite abelian categories. The **Deligne tensor product**, denoted by $\mathcal{C} \boxtimes \mathcal{D}$, is a k-linear finite abelian category equipped with a k-bilinear functor

$$\boxtimes : \mathcal{C} \times \mathcal{D} \longrightarrow \mathcal{C} \boxtimes \mathcal{D}, \quad (X, Y) \mapsto X \boxtimes Y$$

TODOTOD!!

Definition 2.2.3 (The Category Vec_k)**.** Let k be an algebraically closed field.

- **Objects:** The objects are finite dimensional vectors spaces over k.
- **Morphisms:** Linear maps between vector spaces.

Definition 2.2.4 (The Category $\mathrm{Rep}_k(G)$)**.** Let G be a group and k a field. The *category of linear representations of G over k* consists of the following data:

- **Objects:** The objects $\mathrm{Rep}_k(G)$ are pairs (V, ρ_V), where V is a k-vector space and $\rho_V : G \to \mathrm{GL}(V)$.
- **Morphisms:** For any two objects (V, ρ_V) and (W, ρ_W) in $\mathrm{Rep}_k(G)$, a morphism from V to W is a k-linear transformation $f : V \to W$ that is G-invariant (also called an *intertwiner*). That is, for every element $g \in G$, the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "\rho_V(g)"] \arrow[d, "f"'] & V \arrow[d, "f"] \\ W \arrow[r, "\rho_W(g)"'] & W \end{tikzcd}$$

  In algebraic terms, f must satisfy the condition:

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad \forall g \in G$$

  The set of all such morphisms is denoted by $\mathrm{Hom}_G(V, W)$.

We can further endow $Rep_k(G)$ with a monoidal structure: For any two objects $(V, \rho_V), (W, \rho_W)$, their tensor product is in $\mathrm{Rep}_k(G)$ by the diagonal action

$$\rho_{(V \otimes W)}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Definition 2.2.5 (Direct product of representations)**.** A representation of $G_1 \times G_2$ is a space where the pair (g_1, g_2) acts linearly. SO this is a pair

$$(V, \rho), \rho : (G_1 \times G_2) \to GL(V)$$

Now, let $(\mathrm{Rep}_{\mathbb{C}}(G), \otimes, \mathbb{C})$ be the monoidal category of representations. We equip the category with a tensor product that acts as follows:

TODO....

Theorem 2.2.6 (Fusion Representation Category of finite groups)**.** Let G be a finite group. Then $\mathrm{Rep}(G)$ is a fusion category.

Theorem 2.2.7. Let G_1 and G_2 be finite groups and let $\mathrm{Rep}(G_1), \mathrm{Rep}(G_2)$ be their respective symmetry categories. Then there is an equivalence of categories

$$\mathrm{Rep}(G_1 \times G_2) \simeq \mathrm{Rep}(G_1) \boxtimes \mathrm{Rep}(G_2)$$

Big issue here: Rep(SU(2)) is not a fusion category as it is not finite. But if i read it correctly, the deligne product only requires abelianness of its groups. Now, $\mathrm{Rep}(G)$ is monoidal for any G but is it abelian?

So this is actually true, it is abelian. We could obtain a restricted versions by for example always restricting $SU(2)$ to some $SU(2)_k$ for some finite k. But i do not yet know how to present this nicely.

A second advantage would be that we can always represent fusion trees as monoidal categories (is this true or is fusion required here again?), and we can write down a unified theory for the fusion trees. Then I hope to write a nice method to view the "eval" of such a fusion tree, i.e. its clebsch gordan tensors by contracting down the topology of the tree.

2.3 Categories Needed

The motivation for this section is to introduce the necessary categoy theory background, and provide all the necessary theorems for a format that includes both representations and fusion categories.

Definition 2.3.1 (Semisimple Category)**.** An abelian category C is semisimple if every object is either simple or a direct sum of simple objects. An object is *simple* if its only subobjects are 0 or X itself. Any object can be written as

$$X = \bigoplus_{i \in I} S_i^{\oplus n_i}$$

Lemma 2.3.2 (Schurs 1st lemma (which one?))**.** If S and T are simple objects in a semisimple category over an algebraically closed field (this is always C in our case), then it holds that

$$\mathrm{Hom}(S,T) = \begin{cases} k, & \text{if } S \cong T; \\ 0, & \text{if } S \ncong T. \end{cases}$$

Proof [IR19, Theorem 7.5].

Lets give a coordinatized approach to fusion categories:

Definition 2.3.3 (Fusion Category)**.** A *fusion category* is a finite set of labels (sectors?) L and vector spaces V_c^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \to \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds.

Lemma 2.3.4. Let C be a k-linear monoidal category whose tensor product is k-linear in each variable. Then

$$\bigotimes_{r=1}^{n} \left(\bigoplus_{a_r} X_{r,a_r} \right) \cong \bigoplus_{\hat{a}} \bigotimes_{r=1}^{n} X_{r,a_r},$$

where

$$\hat{a} = (a_1, \ldots, a_n).$$

Lemma 2.3.5 (Finite Groups admit Fusion Category)**.** Let G be a finite group, the $\mathrm{Rep}(G)$ is a fusion category.

Let $\mathrm{Vec} \coloneqq (\mathrm{FD}\mathbb{C}\mathrm{Vec}, \otimes, \mathbb{C})$ be the monoidal category of finite dimensional complex vector spaces where the last tuple entry $\mathbb{C}$ denotes the unit object, since there are isomorphisms $V \otimes \mathbb{C} \cong V \cong C \otimes V$.

Let $(\mathbb{C}\mathrm{Rep}(G), \otimes, ?)$ denote the representation Category for a compact group G. Here $\otimes$ is the tensor product of representations $(\pi \otimes \rho, V \otimes W)$.

Definition 2.3.6 (Direct sum)**.** Given a category C with objects $\{a_n\}$, an object $\bigoplus_{i=1}^n a_i \in C$ is called the *direct sum.* This is equipped with the injections and projections

$$\iota_i : a_i \to \bigoplus a_i \quad \text{and} \quad \pi_i : \bigoplus a_i \to a_i$$

such that

$$\pi_i \circ \iota_i = id_{a_i} \forall i$$

Lemma 2.3.7 (Peter Weyl Theorem)**.** Let G be a compact group. Then all its unitary representations are completely reducible.

2.3.1 Representation Theory

IG remove this since we will only be dealing with finite groups? discuss!

Definition 2.3.8 (Compact Group)**.** A compact group is a group G equipped with a topology such that

$$m : G \times G \to G, \quad (g, h) \mapsto gh,$$

is continuous.

$$i : G \to G, \quad g \mapsto g^{-1}$$

is continuous and G is compact as a topoligical space.

Example 12.

1. Every finite group with the discrete topology is compact.
2. $$U(n) = \{A \in M_n(\mathbb{C}) : A^*A = I\}$$
   is compact for all integer n.
3. All
   $$SU(n) = \{A \in U(n) : \det A = 1\}$$
   are compact.

Definition 2.3.9 (Representation)**.** Let G be a group and V a vector space over a field $\mathbb{F}$ (In our cases $\mathbb{C}$) suffices. A *representation* of G on V is a group homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(V),$$

where $operatornameGL(V)$ denotes the group of invertible linear transformations on V. Equivalently

$$\rho(gh) = \rho(g)\rho(h), \qquad \rho(e) = I, \qquad \rho(g^{-1}) = \rho(g)^{-1}.$$

The vector space V is called the *representation space.*

Definition 2.3.10 (Projective Representation)**.** Given a group G, a linear representation of G over $\mathbb{C}$ of dimension n is a map $U : g \to \mathbb{C}^{n \times n}$.

2.3.2 Fusion Trees

We consider a fusion tree as the basis that spans $Hom(R_a \otimes R_b \ldots \otimes R_m, \otimes R_{a_j} \ldots)$. Since $R_a \otimes R_b$ is semisimple, we have the relationship

$$R_a \otimes R_b \cong \bigoplus_c R_c^{\oplus N_{ab}^c}$$

with embedding (make this formal)

$$g_{c,\mu}^{ab} : R_c \to R_a \otimes R_b$$

and projections

$$f_{ab}^{c,\mu} : R_a \otimes R_b \to R_c$$

$$id_{R_a \otimes R_b} = \sum_{c \in L} \sum_{\mu=1}^{N_{ab}^c} g_{c,\mu}^{ab} f_{ab}^{c,\mu}$$

Lemma 2.3.11.

$$Hom(R_{a_1} \otimes \cdots \otimes R_{a_N}, R_b)$$

is spanned by ordinary fusion trees.

Proof. Base case: $N = 1$. Nothing to prove here.

INductive case:

Under semisimplicity we get that

$$Hom(R_{a_1} \otimes \cdots \otimes R_{a_n}, R)b) \cong \bigoplus_{c,\mu} Hom(R_c \otimes R_{a_3} \otimes \cdots \otimes R_{a_n}, R_b)$$

$\square$

2.3.3 Monoidal data

Definition 2.3.12 (Rep($SU(2)$))**.** Let (V, ρ_V) be a representation.

Definition 2.3.13 (Twists from R moves)**.** Suppose we are in a braided monoidal category with F and R symbols. Then we compute the *twist* (a self braiding) as follows:

$$R_c^{ab} R_c^{ba} = \frac{\theta_c}{\theta_a \theta_b}.$$

(state more possibilities here.)

Example 13. Take $sVect$. Our simples are 1 and f. This *braiding* is in fact symmetric. Thus we have that

$$1 = \frac{\theta_1}{\theta_f^2} =$$

2.4 Anyonic Hamiltionians

Symmetric States

A state $|\psi\rangle$ is symmetric if $U^n(g)\psi = \psi$. Similarly, its densitiy need require that $U^n(g)\rho U^n(g) = \rho$.

2.4.1 1D Lattice Models

So from what i can read here (also the nlab or wikipedia) a spin chain specifically only refers to a per site construction with a tensor product hilbert space. An anyon chain is, well, an anyonic chain. These do not admit such a decomposition. So we should call these simply 1D lattice models.

In [Fei+07], anyonic chains, specifically in the Fib model, are introduced simply as the fusion space of n anyons. Further, the authors define a primitive hamiltonian that assigns -1 to the interaction $\tau \otimes \tau = 1$ and 0 to the interaction $\tau \otimes \tau = \tau$. This is in turn realized on the basis of the intermediate states, and evaluated using the method of exact diagonalization [**<empty citation>**]. This is extended in [Tre+08], with longer ranger interaction. It will become apparent that the construction given in these papers is a manualization of the tensor network constructions we will build here. Generalizations to arbitrary models are given in [Gil+13] as well as [BG17]. Generalized spin chains and dualities using module categories are given in [Eck25].

The tensor network formalism is handled in [Pfe+10], [Pfe12] and [Aye17], and in terms of categories this is done in [Unf24] and [DH25].

For the golden chain, we consider $n+2$ Fib anyons as our ground state. Their combined state yields a Fusion Tree Hom$(\tau^{\otimes n+1}, \tau)$ with intermediate charges $x_1, \ldots, x_n$. These intermediate charges are further restricted as in that there are no possibilities to assign adjacent 1s to them (In fact this is what causes the dimension of the hilbert space to reduce to ϕ^n).

Generally, we can define an analogue of a general state of this N site lattice.

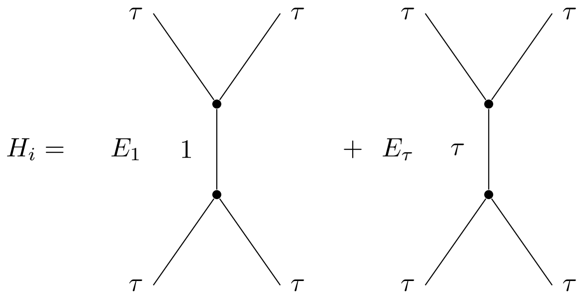
There is a three local hamiltonian for the golden chain defined as

$$H_i = -\sum_{x_i'} [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{x_i}^{x_i'} E_{x_i'} [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{x_{i'}}^{x_{i''}}$$

Write this three local hamiltonian in an operator form

$$H_i = -J \begin{pmatrix} 1 & & & & \\ & 0 & & & \\ & & 0 & & \\ & & & \varphi^{-2} & \varphi^{-\frac{3}{2}} \\ & & & \varphi^{-\frac{3}{2}} & \varphi^{-1} \end{pmatrix}$$

We can build this as an anyonic operator in $Hom(\tau \otimes \tau, \tau \otimes \tau)$ as



.

In this case we can view the golden chain approach chosen above as a manualization of the tensor network process.

We can also write H_i in its operator form matrix as

$$H_i \in \bigoplus_{a \in 1,\tau} Hom(\mathbb{C}, \mathbb{C})$$

as

$$H_i = -J \quad \begin{array}{c} \\ 1 \\ \tau \end{array} \begin{array}{c} \begin{array}{cc} 1 & \tau \end{array} \\ \begin{pmatrix} E_1 & 0 \\ 0 & E_\tau \end{pmatrix} \end{array}$$

$$H_i = -J \quad \begin{array}{c|ccccc} & |1\tau1\rangle & |\tau\tau1\rangle & |1\tau\tau\rangle_1 & |\tau1\tau\rangle_2 & |\tau\tau,\tau\rangle_3 \\ \hline |x_0, \mathbf{1}, \tau\rangle & 1 & & & & \\ |x_0, \tau, \mathbf{1}\rangle & & 0 & & & \\ |x_0, \tau, \tau\rangle_1 & & & 0 & & \\ |x_0, \tau, \tau\rangle_2 & & & & \varphi^{-2} & \varphi^{-\frac{3}{2}} \\ |x_0, \tau, \tau\rangle_3 & & & & \varphi^{-\frac{3}{2}} & \varphi^{-1} \end{array}$$

2.4.2 Anyonic States

For simple lattice states, [Sin+14] defines a bipartition of the lattice into anyons A and B. Then we consider the whole space under charge a_{tot} as

$$V_{a_{tot}} = \bigoplus_{N_{ab}^{a_{tot}}=1} (V_a^A \otimes V_b^B)$$

What i dont get here is why we are allowed to tensor both these sectors together without the space $V_{ab}^{a_{tot}}$ explicitely.

TODO delete all of this and introduce this over the Hom category.

Anyhow, the fundamental building blocks we are using are the vector spaces V_c^{ab} (where we refer to this as the fusion space from types a and b to c). This allows us to build the vector space of two anyonic types a, b as the sum over the fusion outcome:

$$V^{ab} = \bigoplus_{c \in Irr(C)} V_c^{ab}$$

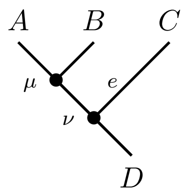
We can directly introduce the R move (a braiding of the initial anyons) to obtain the isomorphic space V^{ba} as

$$R : \bigoplus_{c \in Irr(C)} V_c^{ab} \cong \bigoplus_{c \in Irr(C)} V_c^{ba}$$

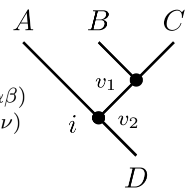
We can also write the space $V_c^{ab} = Hom(a \otimes b, c)$ in the categorical language. Here we also need to realize that there is a difference between the isomorphism β and the action of R, which is really on the hom space $\mathrm{Hom}(-, c)$. So this would be a yoneda perspective? Since we know R for all $a, b, c \in C$, we also know the action of R as an isomorphism. Well technically i believe that this allows us to determine for all simples c, and then semisimplicity extends this to all subjects (this would be nice to write down) and then the yoneda embedding is enough to know the whole space. Similarly, in our Rep(G) categories (which i actually belive to be symmetric? Find this in etingof)

I think we may be able to build a nice summary with this diagram:

$$\begin{tikzcd}
(A \otimes B) \otimes C \arrow[rr, "\underset{\sim}{\alpha_{ABC}}"] \arrow[d, "Hom(-,D)"] & & A \otimes (B \otimes C) \arrow[d, "Hom(-,D)"] \\
Hom((A \otimes B) \otimes C, D) \arrow[rr, "\underset{\sim}{\alpha_{A,B,C}}", "\text{associator induces isomorphims under yoneda}"'] \arrow[d, phantom, "\cong"', "\text{semisimplicity}"] & & Hom(A \otimes (B \otimes C), D) \arrow[d, phantom, "\cong"', "\text{semisimplicity}"] \\
\bigoplus_{c \in Irr(CAT)} Hom(A \otimes B, c) \otimes Hom(c \otimes C, D) \arrow[rr, phantom, "\cong"] \arrow[d, phantom, "="] & & \bigoplus_{d \in Irr(\mathsf{Cat})} Hom(A \otimes d, D) \otimes Hom(B \otimes C, d) \arrow[d, phantom, "="] \\
\bigoplus_c V_c^{AB} \otimes V_D^{cC} \arrow[rr, phantom, "\cong"] \arrow[d, phantom, "\rotatebox{90}{$\in$}"] & & \bigoplus_c V_D^{Ac} \otimes V_c^{BC} \arrow[d, phantom, "\rotatebox{90}{$\in$}"] \\
\underset{\mu \in 1,\ldots,N_c^{AB}}{|(A,B);c;\mu\rangle} \otimes \underset{\nu \in 1,\ldots,N^{cC,D}}{|c,C;D;\nu\rangle} \arrow[rr, "F_D^{ABC}", "\text{6j-symbols}"'] & & \sum_{f,\alpha,\beta} [F_D^{ABC}]^{(f\alpha\beta)}_{(c\mu\nu)} |A,f;D;\beta\rangle \otimes |B,C;f;\alpha\rangle
\end{tikzcd}$$



$$= \sum_{f,\alpha,\beta} [F_D^{ABC}]^{(f\alpha\beta)}_{(e\mu\nu)}$$



2.4.3 Rydberg-blockade ladder

This is given by the fusion category $so(3)_2$. Here we have fusion rules

$$1 \otimes 1 = 0 \oplus 1 \oplus 2, \quad 2 \otimes 1 = 1, \quad 2 \otimes 2 = 0.$$

2.5 I dont know where to put this yet

If we start out on anyonic chains, like the $SU(2)_3$ case, this provideds a nice generalization for chains with $C = Rep(SU(2)_k)$ with fusion rules

$$j_1 \times j_2 = \sum_{j=|j_1-j_2|}^{\min(j_1+j_2, k-j_1-j_2)} j.$$

However now, if we approach this in the case of $SU(2)_\infty$ these fusion rules read

$$j_1 \times j_2 = \sum_{j=|j_1-j_2|}^{j_1+j_2}$$

Is this the same as the Q deformations? It is for $q = 1$ and $k = \infty$ but what about the intermediates?

2.6 MPO-symmetries

So im not sure how this works completely but there seems to be way more literature on this than on (globally) symmetric tensors. However this also seems to be connected to anyonic networks. Also to the Category of MPS and MPO?

There is an account of (projective) representation on a peps that act on all levels in [Bri14]. Actually there are more such as chapter IV section 2 of [CWV26], [Wil+16]

Some sources here: On Lattice Models with Module Categories: [Eck25], [Loo+23] [LDV24b] [LDV25]. Also chapters IV and V of [CWV26]. On the structure see [Liu+25]

I think the original motivation for this is to have the following constraint on MPOs:

$$MPO_a \circ MPO_b = \sum_c N_{ab}^c MPO_c$$

which we can obtain by a fusion tensor X_{ab}^c that obeys an intertwining property.

We immediately obtain our case of global symmetry by encoding

$$\rho(g) = \rho_1(g) \otimes \cdots \otimes \rho_n(g)$$

as a MPO with bond dimension 1 by setting

$$W_{11,ij}^{[a]} = \rho_a(g)_{ij}$$

Call $\rho(g) = MPO_g$. Then $\rho(g)\rho(h) = \rho(hg)$

If we want $MPO_g MPO_h = MPO_{hg}$ we also have MPO as a unitary representation with bond dimensions > 1. So this generalizes the symmetry approach.

TODO: check if we can give symmetries such as in the wanniers kramer dual that are given by $Z_i Z_{i+1}$ by not a $Rep(G)$ but instead a $Rep(\mathrm{G})$ where G is a groupoid.

Definition 2.6.1 (2-Category of MPS)**.** Let A and B be arbitrary MPS with bond collections d^A, d^B. Now, define the 1-morphism category $Hom(A \to B)$ where the objects are MPO operators M together with a reduction isometry Y.

2.7 Appendix

Implementations that are available:

Name	**Symmetry**	**Citation**(Software, Publication)	Language
ITensor	Nonabelian (Fermion?)	[FWS22]	Julia
Cyten	Categorical	[UH] [Unf24]	python
TeNPy	Abelian	[Hau+24]	python
QSpace	Nonabelian	[Wei24] [Wei12]	?
ChemTensor	$SU(2)$	[Men26]	C
UniTensor	?	[<**empty citation**>]	? (Python, C++)
YasTn	Abelian	[Ram+25]	Python
TensorKit.jl	Categorical	[DH25]	Julia
Cytnx	Abelian only (i think)	[<**empty citation**>]	C++
quimb	Fermion?	[<**empty citation**>]	julia
MPSKit.jl	like TensorKit?	[<**empty citation**>]	julia
TeMFpy	Fermion?	[HS25]	python
abeliantensor	$U(1)$ and $\mathbb{Z}_n$	[Hau26]	python
symmray	Fermion	[Gao+25] [Gra26]	?
grassmanntn	Grassmann	[Yos26]	?
SymLibrary	? (OnSite)	[McM19] [MSB20]	MatLab?

Table 2.1: Overview of tensor network libraries and symmetry support. (There are some closed source things here as well, also what is the software used by singh et al.?). Software is included if it has a website or documentation and an account of the symmetries it exploits. Applications or theory are listed in the Citation column.

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